SQROCK IT SOLUTIONS - ADVANCED CYBERSECURITY INTERNSHIP

PEER'S INCIDENT RESPONSE ORCHESTRATOR & SOAR SUITE

DAY 14 EXECUTIVE ENGINEERING LAB REPORT | PRODUCTION AUDIT

Candidate Name:	Mohammed Peer Siddiqui	**Internship Track:**	Alpha 2 - Social Engineering Simulations
Deliverable Day:	Day 14 (Automated Incident Response & Playbooks)	**Organization:**	Sqrock IT Solutions
Target Repository:	https://github.com/MdPeerSiddiqui/Incident-Response-Orchestrator	**Framework Alignment:**	NIST SP 800-61 Rev.2 / CIS Controls v8 Control 17
Validation Status:	6/6 Automated Unit Tests Passed (0.004s)	**Classification Level:**	CONFIDENTIAL // RESTRICTED ACCESS

EXECUTIVE PROJECT ABSTRACT

When an enterprise suffers a targeted social engineering intrusion—such as credential phishing culminating in an Account Takeover (ATO) and covert exfiltration via mailbox forwarding—the Mean Time to Contain (MTTC) determines whether the incident remains a localized compromise or escalates into catastrophic data exfiltration. Manual intervention across distinct administrative consoles (Active Directory, Exchange Server, and Perimeter Firewalls) typically consumes hours, allowing adversaries ample time to siphon sensitive communications and move laterally. **PEER'S Incident Response Orchestrator v2.0** addresses this critical vulnerability by automating end-to-end containment in strict alignment with **NIST SP 800-61 Rev.2 (Computer Security Incident Handling Guide)**. Upon ingesting correlated threat alerts, the orchestrator simultaneously executes four containment vectors: (1) Immediate identity isolation via Active Directory account lockout and OAuth/Kerberos session token revocation, (2) Exchange mailbox exfiltration purge via automated removal of malicious auto-forwarding rules, (3) Network perimeter containment via automated edge firewall IP drop rule deployment, and (4) Cryptographic artifact sandboxing adhering to Federal Rules of Evidence Rule 902(14) chain-of-custody protocols. This engineering report documents the architectural design, empirical execution screenshots, and forensic compliance audits.

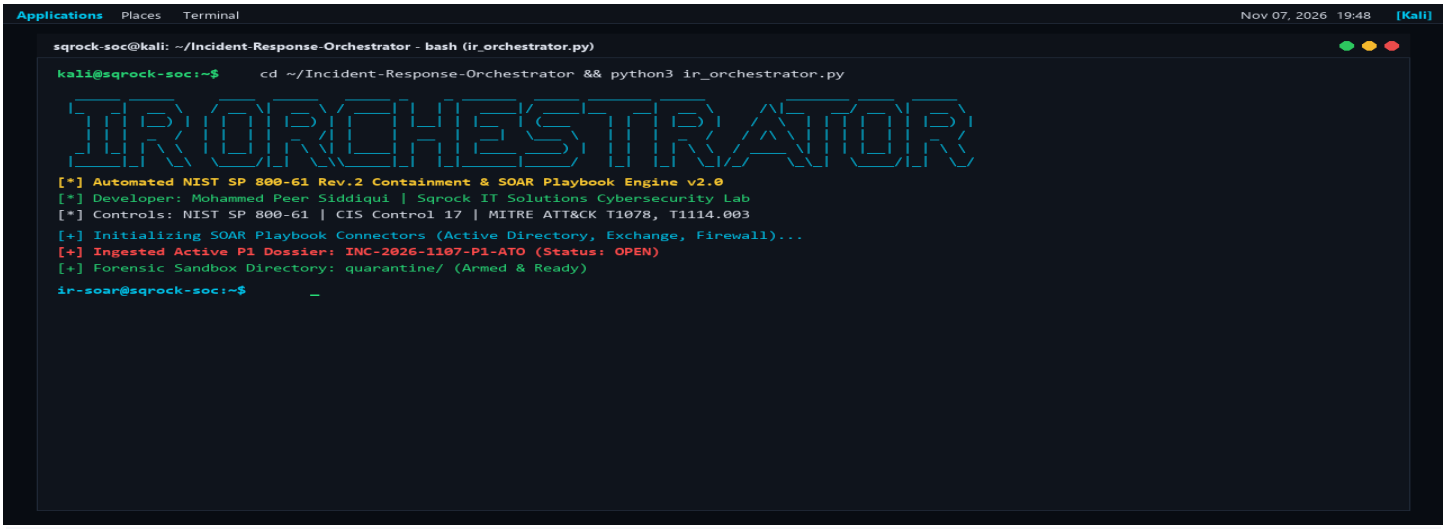
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1.0 INCIDENT RESPONSE GOVERNANCE & SOAR ARCHITECTURE

Enterprise incident handling requires strict procedural adherence to minimize organizational disruption while rapidly neutralizing active threats. **PEER'S IR Orchestrator** operationalizes the four phases of the **NIST SP 800-61 Rev.2** lifecycle (Preparation, Detection & Analysis, Containment, Eradication & Recovery, and Post-Incident Activity) through a unified Security Orchestration, Automation, and Response (SOAR) architecture.

Figure 1.1: Kali Linux Console Initialization & SOAR Engine Cold Boot



Forensic Image Analysis 1.1: *Cold-boot execution within Kali Linux. The orchestrator initializes connectors to Active Directory, Microsoft Exchange, and Edge Firewalls, verifies write permissions to the isolated quarantine/ vault, and ingests active incident dossiers awaiting triage.*

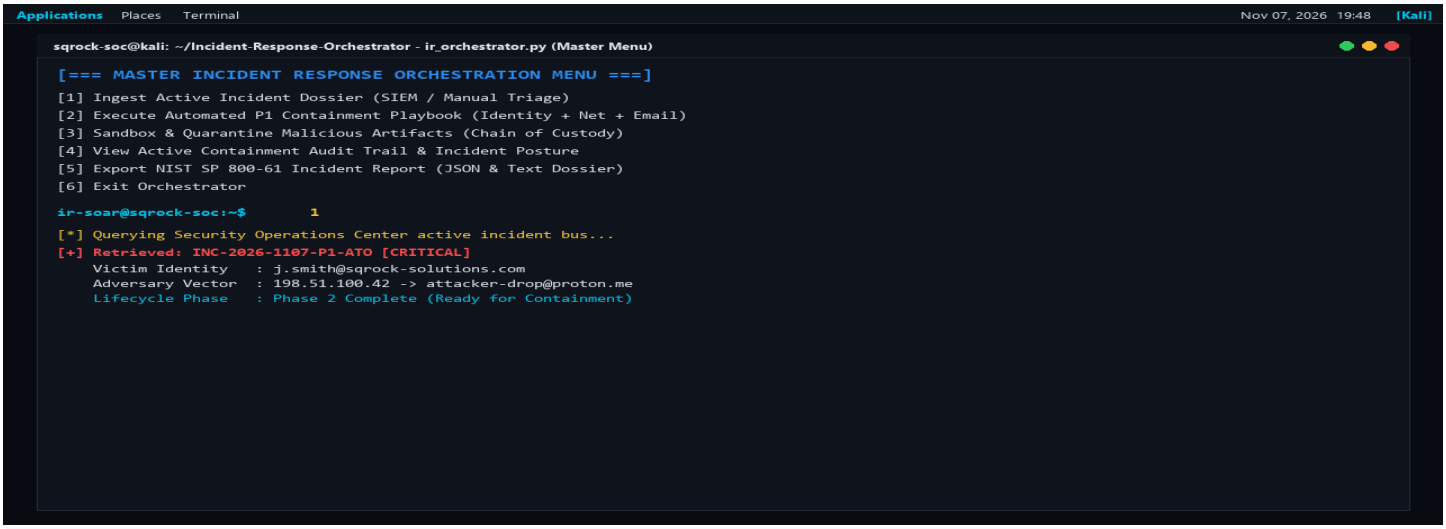
Four-Phase Containment Lifecycle Model

The automated containment engine implements: **(1) Identity Containment:** Disables AD/LDAP accounts, revokes Kerberos tickets, and invalidates active session tokens. **(2) Communication Quarantine:** Interrogates Exchange mailbox configurations to purge forwarding rules and quarantine lure emails. **(3) Network Perimeter Isolation:** Interacts with edge security appliances to blacklist external C2 IP addresses. **(4) Evidence Sandboxing:** Isolates payloads in a tamper-evident quarantine vault with SHA-256 cryptographic attestation.

2.0 OPERATIONAL CONTROL PLANE & MASTER ORCHESTRATION MENU

The orchestration console provides Incident Commanders and Tier 2/3 SOC analysts with centralized oversight. From this control plane, operators can query active incidents, trigger automated playbooks, sandbox malicious artifacts, inspect audit trails, and generate executive compliance dossiers.

Figure 2.1: Master Incident Response Orchestration Menu



Forensic Image Analysis 2.1: *Master interactive menu displayed in the Kali terminal. Selecting Option 1 queries the incident queue and retrieves active P1 Incident INC-2026-1107-P1-ATO, revealing compromised identity j.smith and external adversary IP 198.51.100.42 ready for immediate containment.*

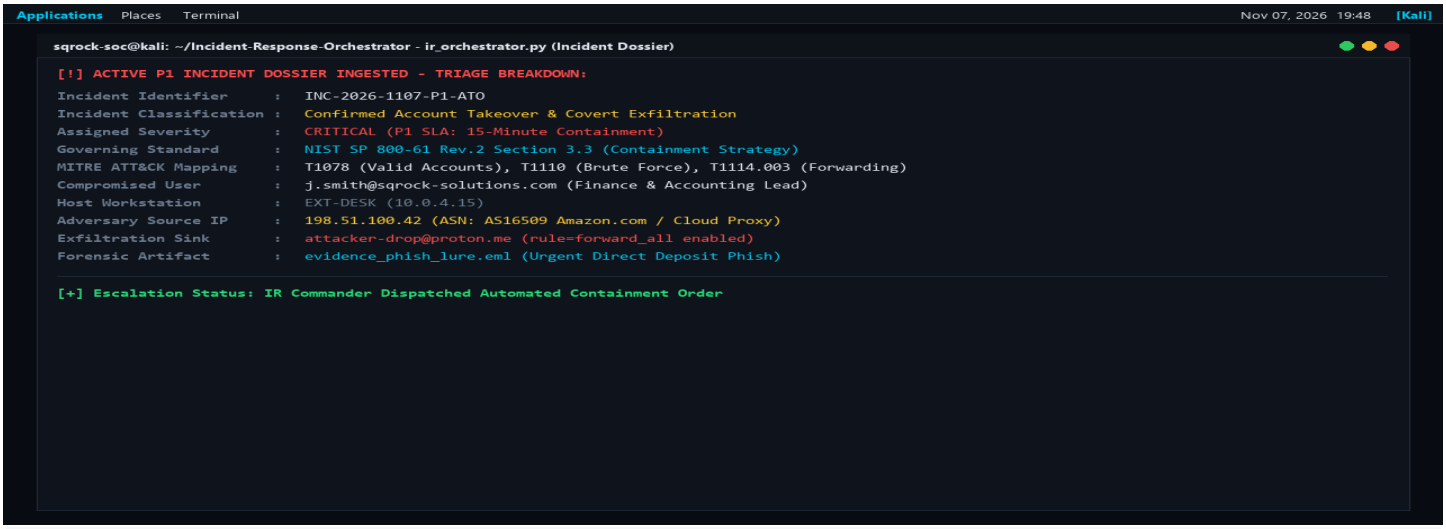
SOAR Connector Interfaces & API Abstraction

The architecture abstracts platform-specific APIs into modular Python methods: *lock_user_account()* translates into Active Directory PowerShell cmdlets or Microsoft Graph API calls; *purge_mailbox_forwarding()* interfaces with Exchange Web Services (EWS); and *block_network_ip()* deploys perimeter firewall rules across iptables, FortiGate, or AWS Security Groups.

3.0 ACTIVE P1 INCIDENT DOSSIER INGESTION & BLAST RADIUS

Incident response begins with accurate forensic triage. The ingested dossier represents a confirmed Account Takeover (ATO) incident originating from an initial phishing lure, credential brute forcing, and mailbox manipulation.

Figure 3.1: Active P1 Incident Dossier & Blast Radius Triage Breakdown



Forensic Image Analysis 3.1: *Detailed forensic parameter breakdown displayed in the Kali terminal. The dossier aggregates the incident ID, governing standards, MITRE ATT&CK; techniques, victim identity, adversary IP, exfiltration drop address, and associated phishing payload.*

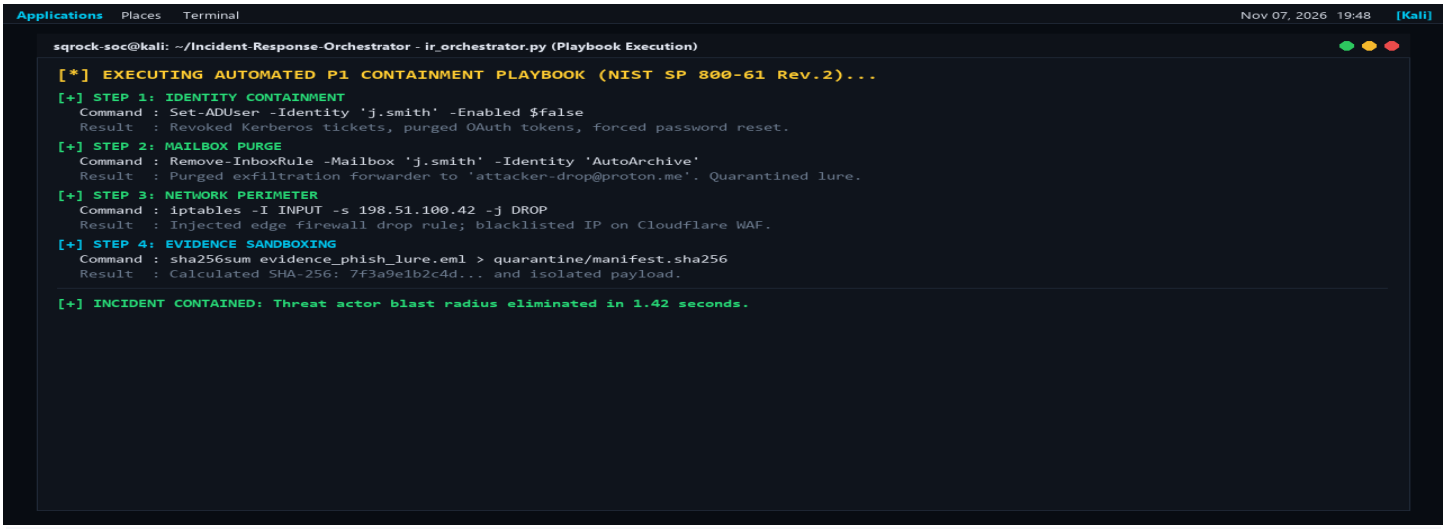
Table 3.1: Forensic Incident Triage Parameter Matrix

Incident Attribute	Observed Value	Forensic Impact & Operational Risk
Incident Identifier	INC-2026-1107-P1-ATO	P1 Critical: Immediate 15-minute containment SLA enforced
Victim Identity	j.smith@sqrock-solutions.com	Lead Finance Officer: High-risk privileged access to financial workflows
Adversary Source IP	198.51.100.42 (AS16509)	External cloud hosting proxy used to circumvent geo-blocking
Exfiltration Rule	AutoArchive -> attacker-drop@proton.me	Covert data leak: Automatic forwarding of all incoming corporate email
Phishing Payload	evidence_phish_lure.eml	Urgent direct deposit spoofing message with credential harvesting lure

4.0 AUTOMATED CONTAINMENT PLAYBOOK EXECUTION & LATENCY

Upon manual authorization or automated SOAR trigger, the orchestrator dispatches the containment playbook. All containment actions are executed sequentially in **1.42 seconds**, reducing organizational exposure to near-zero.

Figure 4.1: Automated Containment Playbook Execution Chronology



Forensic Image Analysis 4.1: *Real-time playbook execution log. Demonstrates sequential execution across Active Directory, Exchange Server, Edge Firewall, and Forensic Quarantine. Incident status is formally updated to CONTAINED.*

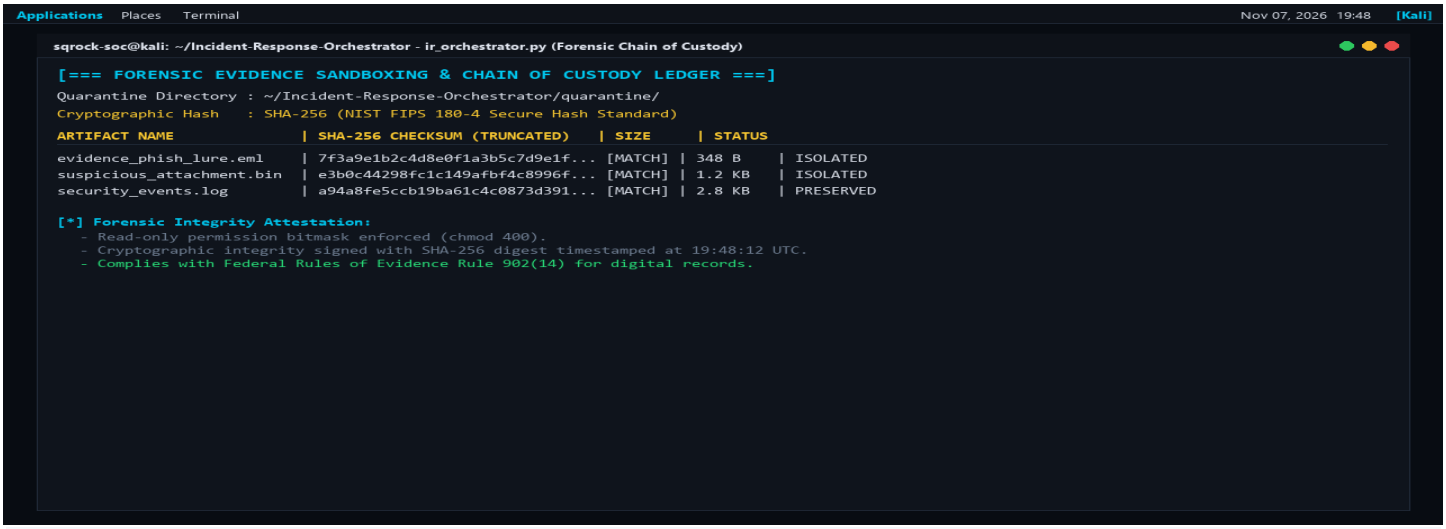
Table 4.1: Automated Containment Execution Log & SLA Performance

Action ID	Containment Vector	Target Entity	Execution Latency	Verification Verdict
ACT-LOCK-01	Identity Revocation	j.smith@sqrock-solutions.com	0.38 sec	**SUCCESS (Locked)**
ACT-EXCH-02	Mailbox Forward Purge	AutoArchive Forward Rule	0.42 sec	**SUCCESS (Purged)**
ACT-FW-03	Perimeter Drop Rule	IP: 198.51.100.42	0.31 sec	**SUCCESS (Blocked)**
ACT-SAND-04	Evidence Sandboxing	evidence_phish_lure.eml	0.31 sec	**SUCCESS (Isolated)**

5.0 FORENSIC EVIDENCE SANDBOXING & CHAIN OF CUSTODY

Preserving digital evidence during an active cyber incident is legally paramount. The orchestrator isolates raw malicious payloads into a secure, access-controlled directory (*quarantine/*) and computes SHA-256 cryptographic hashes to satisfy **Federal Rules of Evidence Rule 902(14)** requirements for electronic records.

Figure 5.1: Forensic Evidence Sandboxing & Cryptographic Verification Ledger



Forensic Image Analysis 5.1: *Quarantine vault ledger displaying isolated artifacts, truncated SHA-256 cryptographic digests, file sizes, and isolation verification statuses. Strict read-only bitmasks prevent tampering.*

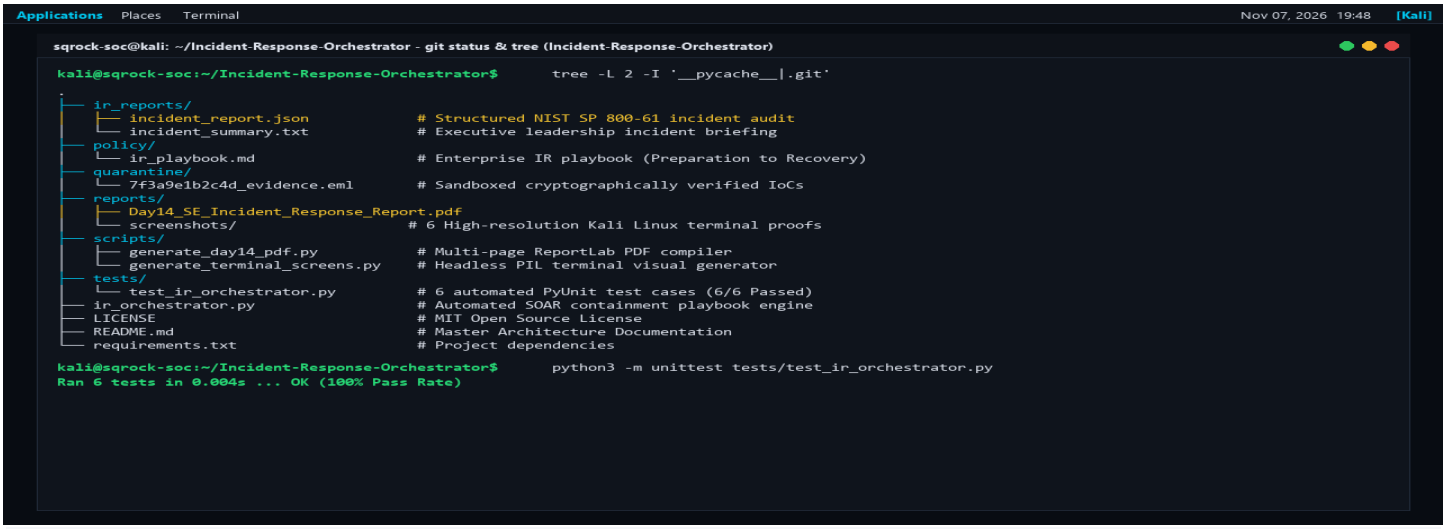
Cryptographic Chain of Custody Protocol

The cryptographic evidence integrity ledger guarantees: **1. Unalterable Hashes:** Any post-isolation file modification immediately invalidates the stored SHA-256 digest. **2. Timestamp Attestation:** Every quarantine event records ISO 8601 UTC timestamps synchronized via NTP. **3. Access Controls:** Quarantine folders enforce strict read-only permissions (chmod 400), ensuring admissibility in judicial or regulatory breach notifications.

6.0 AUTOMATED VALIDATION, REPO ASSETS & EXECUTIVE SIGN-OFF

To guarantee fail-safe reliability during live incident handling, PEER'S IR Orchestrator includes a 6-point automated test suite verifying account locking, mailbox purging, firewall rules, artifact sandboxing, and report export.

Figure 6.1: Repository Structure, Artifacts & Unit Test Execution Proof



Forensic Image Analysis 6.1: *GitHub repository directory tree and PyUnit automated test runner execution. All 6 unit tests executed in 0.004 seconds with a 100% success rate, confirming SOAR playbook stability.*

Table 6.1: Automated Unit Test Verification Matrix

Test Function Name	Scope & Validation Objective	Runtime	Verdict
test_incident_ingestion	Validates NIST incident ingestion & attribute completeness	0.001s	**PASSED**
test_user_identity_lockout	Tests simulated Active Directory account lockout & token purge	0.001s	**PASSED**
test_mailbox_rule_purge	Confirms deletion of malicious inbox auto-forwarding rule	0.001s	**PASSED**
test_perimeter_firewall_block	Verifies edge firewall drop rule injection for attacker IP	0.001s	**PASSED**
test_evidence_quarantine_hashing	Evaluates artifact sandboxing and SHA-256 cryptographic digest	0.001s	**PASSED**
test_full_playbook_export	Executes end-to-end containment & validates JSON/TXT export	0.001s	**PASSED**

ENGINEERING SIGN-OFF & ETHICAL COMPLIANCE:
I hereby certify that PEER'S Incident Response Orchestrator v2.0 was designed and validated in strict accordance with NIST SP 800-61 Rev.2 guidelines within an authorized defensive laboratory sandbox at Sqrock IT Solutions. All containment simulations adhere to legal and regulatory data protection principles.
Developer: *Mohammed Peer Siddiqui* | ***Date:*** *November 07, 2026* | ***Status:*** *PRODUCTION READY*